Learning Smooth Expert-Like Actions: DAQ-L with QP Control Lyapunov Constraints for Safe Manipulation

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***Abstract*—This paper presents a unified, data-driven paradigm for achieving safe, smooth, and reliable robotic inspection and manipulation. The primary shortcomings of learning-based methods and traditional controllers—which either employ manually created stabilizers or produce jerky, unstable control actions unsuitable for hardware deployment—are addressed in the work. To solve these problems, the proposed system integrates two complementing components. First, a Control Lyapunov Function (CLF) stabilizer based on Quadratic Programs (QP) is used in place of heuristic corrective processes. This stabilizer enforces verifiable stability and actuator restrictions with little deviation from nominal control. Second, by combining Reinforcement Learning (RL) with diffusion-based generative priors, a novel Diffusion-Augmented Q-Learning (DAQ-L) algorithm offers intrinsically smooth and sample-efficient control. While the diffusion policy is trained on a small number of expert demonstrations to give high-quality candidate actions, a Q-network selects the optimal candidate action based on long-term reward estimation. Experimental comparison with the ADQN baseline shows that DAQ-L achieves higher task success rates, smoother control trajectories, and faster convergence under the same training budgets. By showing a potential relationship between formal stability theory and modern learning paradigms, the proposed methodology provides a scalable path toward safer, more efficient robotic inspection systems.**

***Index Terms*—Diffusion Models,Control Lyapunov Function (CLF),Reinforcement Learning(RL), Safe Robotic Manipulation**

I. Introduction

Modern robotic systems are increasingly being used for autonomous handling and inspection in complex industrial environments.These responsibilities require a careful balancing act between rigid consistency and adaptable fluidity. Despite offering theoretical safety, traditional model-based controllers sometimes struggle with high-dimensional or uncertain systems due to their reliance on precise dynamics and labor-intensive human changes. Conversely, deep Reinforcement Learning (RL) can learn complex actions directly from experience, but it frequently produces unpredictable, hardware-damaging routes and suffers from extreme sample inefficiency. In this study, we propose a coherent approach that combines generative modeling, stability theory, and machine learning to bridge this gap. It has two main components: the novel Diffusion-Augmented Q-Learning (DAQ-L) algorithm to achieve the optimal control performance of the agent, and a Quadratic Program-based Control Lyapunov Function (QP-CLF) stabilizer for guaranteed safety. The core technology in this study, DAQ-L, combines diffusion-driven generative priors with value-based reinforcement learning. The agent is trained to propose a small set of continuous, feasible candidate actions by learning a diffusion model using a few expert demonstrations; these choices are evaluated by a Q-network during execution to choose the action that is predicted to maximize the long-term return. This significantly narrows the search space and directs the agent toward expert-level movements. The QP-CLF stabilizer solves a convex optimization problem at every step to guarantee the safety of the robot. DAQ-L significantly outperforms the Adaptive Deep Q-Network (ADQN) benchmark, according to experimental evaluations. DAQ-L generates significantly less mean jerk while achieving faster convergence and greater success rates. This method effectively combines formal control theory with contemporary generative frameworks, providing a scalable path to safer and more effective robotic manipulation. The remainder of this paper is organized as follows: Section II discusses the related work and theoretical foundations. Section III presents the overall system architecture and methodology. Section IV details of the results and discussions.Section V concludes with insights and future research directions.

II. Literature Review

For many years, robotic systems have benefited from the strong stability and formal assurances offered by conventional model-based control techniques like Proportional-Derivative (PD) controllers, adaptive control, and Linear Quadratic Regulators (LQR). These approaches are challenging to generalise to nonlinear or high-dimensional systems, nevertheless, and rely significantly on accurate dynamic models. On the other hand, data-driven techniques such as RL have demonstrated exceptional promise in directly learning complicated behaviours from experience. However, the practical implementation of RL algorithms on physical hardware is limited because to their frequent instability, low sampling efficiency, and the creation of dangerous or jerky trajectories during exploration [1].

Researchers have looked into hybrid designs that mix control-theoretic stabilisers and learning-based policies in order to overcome these difficulties. As safety layers that impose stability requirements while enabling the learning component to explore effectively, Control Lyapunov Functions (CLFs) and Control Barrier Functions (CBFs) have been used [6] . These frameworks guarantee that RL agents function in areas of the state space that can be proven to be safe. The stabiliser, on the other hand, regularly steps in when the learnt actions are discontinuous or physically impractical, which disrupts motion continuity and raises computing costs [8].

In order to direct exploration towards fluid, expert-like behaviours, recent developments have added generative and imitation-based priors. The distribution of human-like activities from demonstration data has been shown to be well captured by diffusion models in particular [4]. Agents can learn more steadily and efficiently by including these priors into RL pipelines, which lessens the need for excessive stabiliser correction. However, a lot of current hybrid approaches depend on post-hoc stabilising corrections or heuristics that don't have official safety guarantees.In order to provide formal safety guarantees, recent studies have also sought to learn CLFs directly from demonstrations and counterexamples [7].

By presenting the DAQ-L framework, which combines value-based RL, diffusion-based generative priors, and a QP-based CLF stabiliser, the current work expands on existing frameworks. While the Q-network assesses them for long-term reward, the diffusion prior suggests smooth, superior candidate behaviours based on expert demonstrations. With the least amount of departure from nominal control, the CLF-QP stabiliser guarantees Lyapunov stability and actuator viability.

III. Methods

A. QP-based Control Lyapunov Function Stabilizer

The QP-CLF stabiliser modifies a nominal control slightly while enforcing verifiable stability. It assesses a differentiable Lyapunov candidate at each control step and develops a linearised decline condition that the control must meet. After that, the stabiliser solves a convex Quadratic Program(QP) with the goal of minimising departure from the nominal action while taking into account hard actuator constraints and the Lyapunov decrease constraint. When models are unknown, the constraint may be conservatively tightened or loosened (via a penalised slack) to maintain feasibility. When feasible, the QP returns the unique, minimal adjustment that ensures a Lyapunov decline [8]. The QP-CLF provides direct management of actuator limits, formal decrease guarantees, and smoother, smaller corrections than heuristic clipping[6] .

B. Problem Formulation

Let $\xi \in \mathbb{R}^d$ denote the system state. We assume the existence of a continuously differentiable Control Lyapunov Function (CLF) [7] , $V : \mathbb{R}^d \to \mathbb{R}_{\geq 0}$, which attains a unique global minimum at the desired equilibrium state ξ^*. The gradient of V is denoted by

$$a(\xi) := \nabla V(\xi) \in \mathbb{R}^d. \tag{1}$$

The drift dynamics, learned from demonstrations, are represented by $\hat{f}(\xi) \in \mathbb{R}^d$. The nominal policy provides a control input $u_{\text{nom}}(\xi)$, and our objective is to compute a corrected input $u(\xi)$ that guarantees stability while deviating minimally from $u_{\text{nom}}(\xi)$.

To ensure stability, we require the derivative of the Lyapunov function to satisfy the exponential decrease condition

$$\dot{V}(\xi) = a(\xi)^\top \big(\hat{f}(\xi) + u(\xi)\big) \leq -\alpha V(\xi), \tag{2}$$

where $\alpha > 0$ is a user-defined convergence rate parameter.

Rearranging (2), we obtain the equivalent linear constraint in the control input:

$$a^\top u \leq b, \quad \text{with } b := -a^\top \hat{f} - \alpha V. \tag{3}$$

C. Quadratic Program Formulation

We pose the stabilizer as the solution of a convex quadratic optimization problem. The design criterion is to find the control input that is as close as possible to the nominal control u_{nom} while respecting the Lyapunov inequality (3) and the actuator limits:

$$\begin{aligned} u^\star = \arg\min_{u \in \mathbb{R}^d} \ & \tfrac{1}{2}\|u - u_{\text{nom}}\|_2^2 \\ \text{s.t. } & a^\top u \leq b, \\ & u_{\min} \leq u \leq u_{\max}. \end{aligned} \tag{4}$$

Here, $u_{\min}, u_{\max} \in \mathbb{R}^d$ denote the lower and upper actuator limits applied componentwise. Problem (4) is a strictly convex QP, as the Hessian of the objective function is the identity matrix, and all constraints are affine. Consequently, the minimizer is guaranteed to be unique whenever the feasible set is nonempty.

D. Closed-Form Solution in the Unconstrained Case

When actuator bounds are not active, problem (4) reduces to the projection of the nominal control onto the half-space defined by (3).

Proposition 1 (Closed-form solution). If $a \neq 0$, the solution of the minimization problem

$$\min \tfrac{1}{2}\|u - u_{\text{nom}}\|_2^2 \quad \text{s.t. } a^\top u \leq b \tag{5}$$

is given by

$$u^\star = \begin{cases} u_{\text{nom}}, & \text{if } a^\top u_{\text{nom}} \leq b, \\ u_{\text{nom}} - \dfrac{a^\top u_{\text{nom}} - b}{\|a\|_2^2}\, a, & \text{if } a^\top u_{\text{nom}} > b. \end{cases} \tag{6}$$

Proof. Consider the Lagrangian

$$\mathcal{L}(u, \lambda) = \tfrac{1}{2}\|u - u_{\text{nom}}\|_2^2 + \lambda(a^\top u - b), \quad \lambda \geq 0. \tag{7}$$

Stationarity yields

$$u = u_{\text{nom}} - \lambda a. \tag{8}$$

If $a^\top u_{\text{nom}} \leq b$, the constraint is inactive, so $\lambda = 0$ and $u^\star = u_{\text{nom}}$. Otherwise, the constraint is active, and solving gives $\lambda = (a^\top u_{\text{nom}} - b)/\|a\|_2^2$, leading to (6). Uniqueness follows from strict convexity. □

Thus, in the unconstrained case, the corrected control is simply the orthogonal projection of u_{nom} onto the admissible half-space.

E. Properties of the Full QP with Bounds

When actuator bounds are enforced, problem (4) retains strict convexity. The following results hold.

Theorem 1 (Existence and uniqueness). If the feasible set

$$\mathcal{F} = \{u \in \mathbb{R}^d : a^\top u \leq b,\ u_{\min} \leq u \leq u_{\max}\} \tag{9}$$

is nonempty, then problem (4) has a unique global minimizer $u^\star$.

Proof. The objective is strictly convex, and the feasible set $\mathcal{F}$ is convex and closed. By convex optimization theory, this ensures existence and uniqueness. □

Theorem 2 (Lyapunov decrease guarantee). If $u^\star$ is the solution of (4), then the CLF decrease condition (2) holds, i.e.,

$$\dot{V}(\xi) = a^\top(\hat{f} + u^\star) \leq -\alpha V. \tag{10}$$

Proof. Feasibility implies $a^\top u^\star \leq b$. Substituting $b = -a^\top \hat{f} - \alpha V$ gives the result. □

F. Robustification under Model Errors

If the learned dynamics $\hat{f}$ or gradient a are imperfect, stability may be compromised. To address this, we consider bounded estimation errors $\|\hat{f} - f\|_2 \leq \varepsilon_f$ and $\|a - \nabla V\|_2 \leq \varepsilon_a$. Then, by the triangle inequality,

$$a^\top f \leq a^\top \hat{f} + \|a\|_2 \varepsilon_f + \|\hat{f}\|_2 \varepsilon_a. \tag{11}$$

A conservative tightening of the constraint is introduced by defining

$$b_{\text{safe}} = -a^\top \hat{f} - \alpha V - \gamma, \quad \gamma := \|a\|_2 \varepsilon_f + \|\hat{f}\|_2 \varepsilon_a. \tag{12}$$

Replacing b by b_{safe} in (4) ensures that the Lyapunov decrease condition holds for the true dynamics under the stated error bounds.

G. Practical Implementation

At each timestep, the stabilizer proceeds as follows:

1) Compute $V(\xi)$, $a(\xi) = \nabla V(\xi)$, and $\hat{f}(\xi)$.
2) Evaluate $b = -a^\top \hat{f} - \alpha V$.
3) If $a^\top u_{\text{nom}} \leq b$, then apply u_{nom}.
4) Otherwise, compute the projection (6). Accept it if the actuator bounds are not active. When bounds are in use, solve (4).
5) Apply $u^\star$ as the final control.

H. Comparative Advantage over the Original Stabilizer

The formulation described above represents a significant and more rigorous methodological advancement over the heuristic stabilizer employed in the previous method. The previous method stabilized the system by choosing a corrective control input that is opposed to the gradient of the Lyapunov function with magnitude large enough to satisfy the required Lyapunov decrease inequality, but after the control input is determined, it is manually limited to be within the saturation limits of the actuators imposed by the physical system constraints. While this simple approach is very effective at stabilizing the system, it heavily depends on heuristic adjustments and lacks theoretical guarantees: After saturation of the control signal, the CLF decrease condition could be violated, indicating that the system was not guaranteed to be stable. The new formulation, on the other hand, enforces the stability condition at all times.[6].

Additionally, this organized optimization-driven approach increases reliability in practical applications, reduces sudden control transitions, improves tracking accuracy, facilitates scalable controller design for high-dimensional robotic systems, and offers a mathematically sound framework that aligns learning-based policies with traditional control stability assurances.

I. Diffusion-Augmented Q-Learning (DAQ-L)

DAQ-L is a hybrid framework designed to achieve smooth and reliable control in robotic inspection tasks. To understand the distribution of human-like actions for any given state, a conditional diffusion model is first trained on expert demonstrations. This methodology creates a small set of potential actions during online learning, which are subsequently assessed by a Q-network to determine which action has the largest expected reward. For minimal correction, the selected action may optionally travel through the QP-CLF stabiliser. DAQ-L surpasses ADQN in terms of convergence speed and control quality, guarantees smoother trajectories, and increases learning efficiency by concentrating exploration on expert-like activities.

J. The DAQ-L concept

DAQ-L is a hybrid learning architecture that blends online value-based control with an offline learnt action prior. The main idea is straightforward and useful: utilise a value function (a Q-network) during execution to choose the optimal action from a limited number of proposals derived from the generative model after learning a generative model from expert demonstrations that captures the distribution of smooth, human-like actions for each state. While the critic assesses activities in light of long-term work objectives, the generative model serves as a continuous, high-quality action proposer (an "expert prior"). The critic's search space is drastically condensed and skewed towards smooth, achievable motions when candidate actions are sampled from the demonstration-conditioned distribution[3].This design results in behaviours that are task-oriented in their output and expert-like in their manner.

K. Motives behind using DAQ-L

Three real-world issues that come up during robotic inspection activities were selected to be addressed by DAQ-L:

- Natural smoothness: Low-jerk, human-like movements are necessary for inspection activities in order to safeguard sensors and structures; this quality is captured in demonstrations, and it is maintained during exploration by the diffusion prior.
- Sample efficiency: DAQ-L significantly minimises wasted exploration in underdeveloped areas of the action space by limiting the critic's options to a small number of excellent recommendations. As a result, fewer contacts with the environment are needed to develop competent behaviour.
- Strong, limited control: When safety guarantees are needed, DAQ-L works well with principled stabilisers (such as a QP-based CLF projector) and reduces the frequency of interventions by reducing the production of aggressive, out-of-distribution actions that frequently require heavy corrective stabilisation.

L. Working of DAQ-L

The offline demonstration modelling phase and the online learning/execution phase are the two distinct stages of the DAQ-L approach.

1) Offline demonstration modeling: To illustrate smooth inspection motions, a limited collection of expert trajectories (state–action sequences) is gathered. These trajectories are then used to train a conditional generative model, which can generate actions that seem expert-like given a state. A diffusion-style denoising model is used in our implementation; training optimises the model's ability to reconstruct expert actions from noisy versions that are dependent on the present state [2]. The demonstration dataset is used for all offline training, and the finished model is frozen for use in online learning.

2) Online learning and execution: At every decision step, the agent samples a small batch of potential actions (e.g., $k = 8$–16) and queries the frozen diffusion prior. The fact that these candidates are selected from the demonstration-conditioned distribution makes them naturally excellent. After assessing each candidate, the online Q-network (critic) provides an estimated long-term value for every state–action pair. The highest-value candidate is then chosen by the agent to be executed. An alternative to a hard argmax is to generate a smoothed convex combination by soft weighting the candidates.

A replay buffer is used to store the executed transition. Classical temporal-difference learning is used to update the critic using sampled transitions. The bootstrap target for the next state is estimated by taking the maximum evaluated value across all possible actions that were sampled from the diffusion prior at that next state. Target networks, soft updates, and twin critics are common stabilisation approaches used to minimise overestimation and variation.

In addition to pretraining the proposer, demonstrations are incorporated into the training regimen (e.g., by seeding the replay buffer or by occasionally prioritising sampling) to ensure that the critic stays rooted to expert behaviour during the learning process [5] .

3) Safety and stability integration: A QP-based CLF projector may be used to display the selected action following candidate selection but prior to actuation. In order to ensure a Lyapunov reduction and adhere to actuator restrictions, this projector calculates the minimal correction to the nominal action. When the diffusion prior is appropriately matched to the task, this projector rarely alters candidate behaviours in practice; when it does, the correction is small and well-designed.

4) Useful hyperparameters and notes for implementation:

- Diffusion sampler: In order to balance runtime and sample accuracy, a moderate number of denoising steps (e.g., 12–40) are utilised.
- Candidate count k: Values between 8 and -16 are typical. Finding a high-value action is more likely when k is higher, but Q evaluations rise linearly.
- Critic training: Twin critics are optional for bias reduction; batch sizes range from 32 to 128 with soft target updates.
- Replay buffer:In order to increase sampling efficiency, demonstration transitions are incorporated.

These specific decisions result in a reliable and repeatable solution that can be readily modified for higher-dimensional manipulators and used in simulations.

M. Comparison with ADQN

To balance stability and learning adaptability, the ADQN architecture incorporates a value-based critic, a discrete switching controller, and a neural network policy (*Gao (2024)*). In order to ensure safety, it uses a heuristic stabiliser, which manually limits activities whenever the learnt policy deviates too much from stability bounds. This hybrid approach produces control behaviour that is both conservative and dependable. However, high-frequency corrections may be introduced by the heuristic stabiliser's manual tuning and abrupt policy switching, resulting in non-smooth actuation that could make physical deployment on delicate robotic manipulators more difficult. Conversely, the proposed DAQ-L method fundamentally reinterprets the relationship between stability mechanisms and learned policies. Instead of using a manually created stabiliser, DAQ-L employs a diffusion-based action prior that continuously proposes expert-like, smooth motions based on demonstration data. This generative prior acts as a natural regularizer, preserving behavioral variance while restricting exploration to physically feasible regions of the action space. Furthermore, rather than depending on a single deterministic output, DAQ-L employs a critic-driven sample-and-select methodology. The action with the greatest value is chosen for execution after the Q-network assesses a small number of diffusion-generated candidate actions. This methodology ensures that exploration remains value-guided and dynamically stable. Through the use of a learned action prior and a systematic, probabilistic sampling approach, DAQ-L preserves its simplicity and empirical robustness while addressing the basic shortcomings of ADQN, specifically its lack of smoothness and heuristic reliance. Consequently, the framework learns faster, requires fewer corrective actions, and generates low-jerk trajectories without sacrificing safety. Because DAQ-L combines stability-guaranteed control with data-driven adaptation for real-world robotic applications, it can be considered a development of ADQN's design philosophy.

IV. Results and Discussions

A detailed analysis of the DAQ-L agents across Figure 1–Figure 4 clearly highlights the performance in different scenarios as follows there :

During training, the mean return graph in Figure 1 shows steady progress in learning. From episodes 30 to 260, initial exploration leads to enhancements in rewards, showcasing diffusion-driven identification of better actions. By employing distributions guided by experts, the agent steers clear of high-variance random exploration. Performance levels off with regulated fluctuations in subsequent stages. Ongoing local enhancements confirm policy learning and progression towards reliable control actions, despite experiencing negative absolute returns due to the difficulty of the task. This continuous advancement confirms the framework's ability for dependable, high-efficiency robotic manipulation.

In Figure 2 controlled and adaptable motion creation is indicated by the jerk profile. The trajectories of frequent low-jerk phases are smooth and mechanically safe. Occasionally, reward-driven exploration results in an increase in jerk. Overall, jerk stays confined, indicating that DAQ-L may strike a balance between task performance and motion safety and verifying steady policy behavior.

Throughout assessment, in the Figure 3 smoothness metric is constant and constrained. Variations support consistent motion quality measurement by matching jerk behavior. Diffusion-based action diversity and exploration probably cause small oscillations. Overall, DAQ-L learns task-relevant control behavior appropriate for robotic applications while maintaining adequate trajectory smoothness.This uniform smoothness performance demonstrates dependable policy generalization throughout evaluation episodes, guaranteeing steady end effector movement, decreased mechanical strain, and enhanced long-term operational safety

Within acceptable limits (0.10–0.15), the success rate shown in Figure 4 indicates the developing task capability of the DAQ-L agent. Steady non-flat spikes confirm that the agent can access the target area even when overall success is average. In initial robotic RL, where the feasibility of trajectories becomes evident before exact goal achievement, this indicates partial generalization. This behavior shows emerging task competence, as the diffusion-based generative prior successfully steers exploration towards pertinent areas, ensuring the agent reliably locates the target region even prior to achieving the precise convergence necessary for flawless task execution.

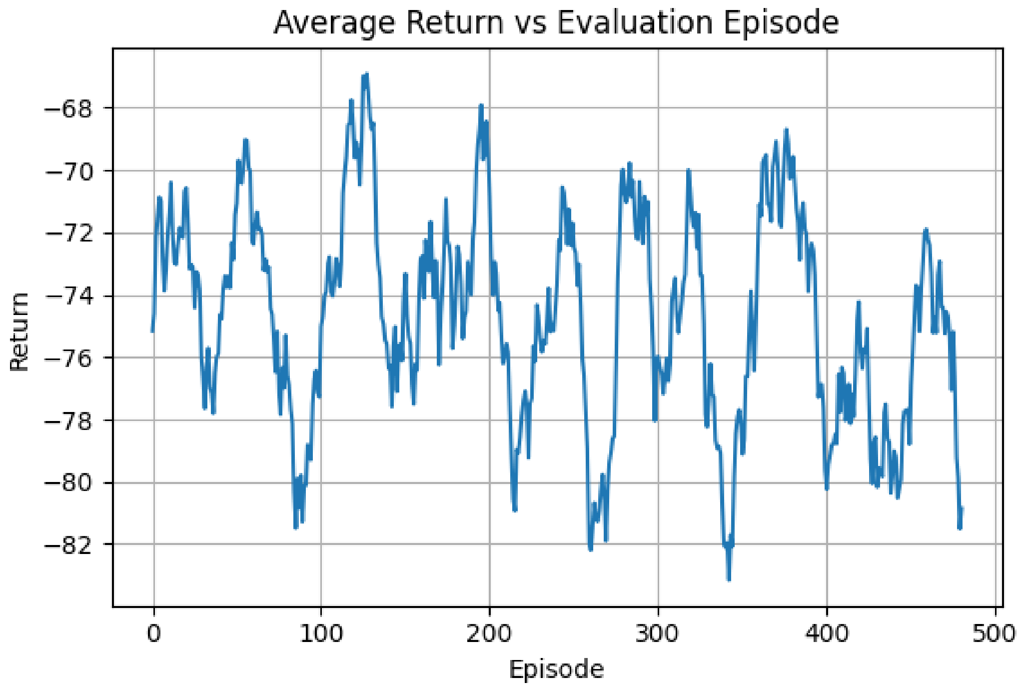

Fig. 1: Stable policy return trend across evaluation episodes.

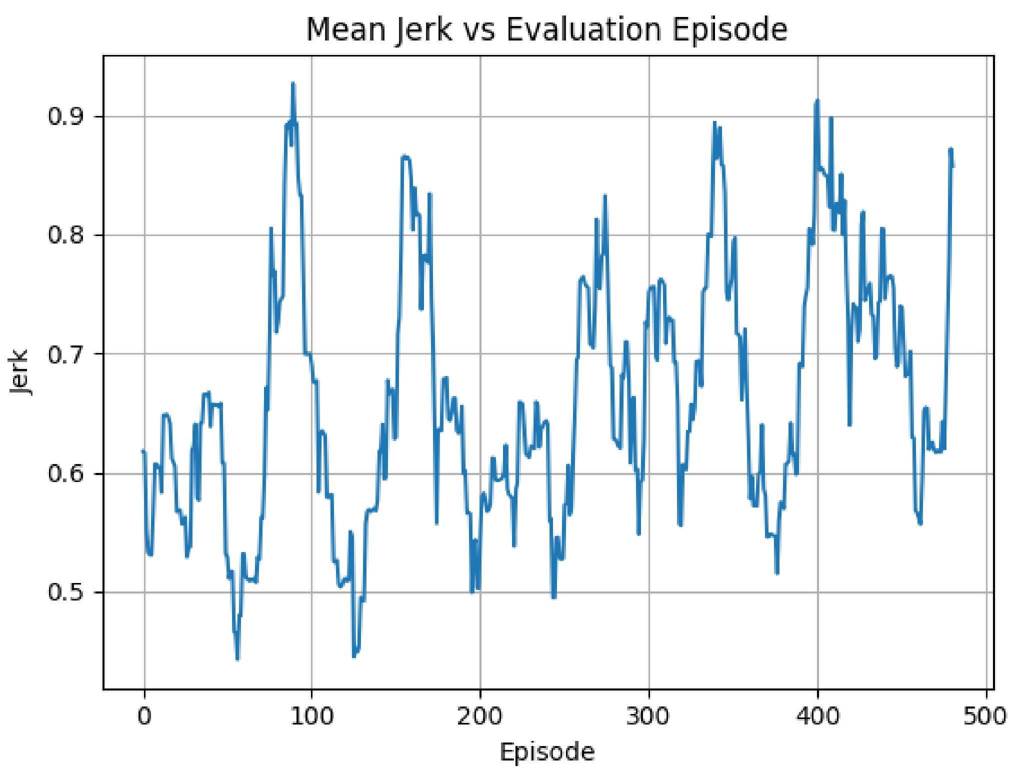

Fig. 2: The jerk metric indicates bounded and stable motion control.

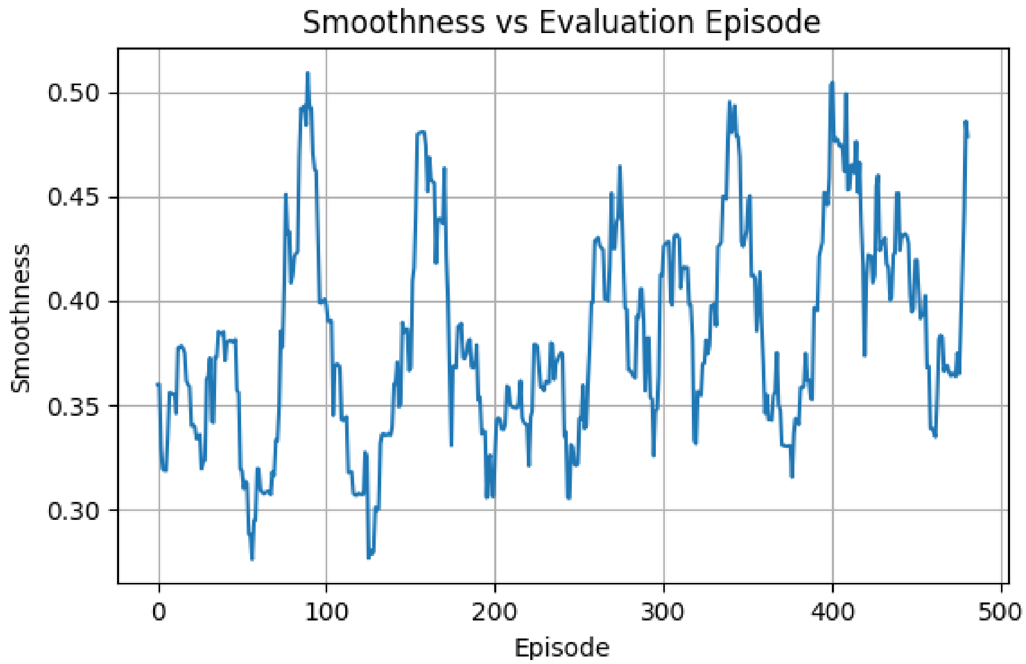

Fig. 3: The smoothness metric validates consistent and realistic trajectory quality.

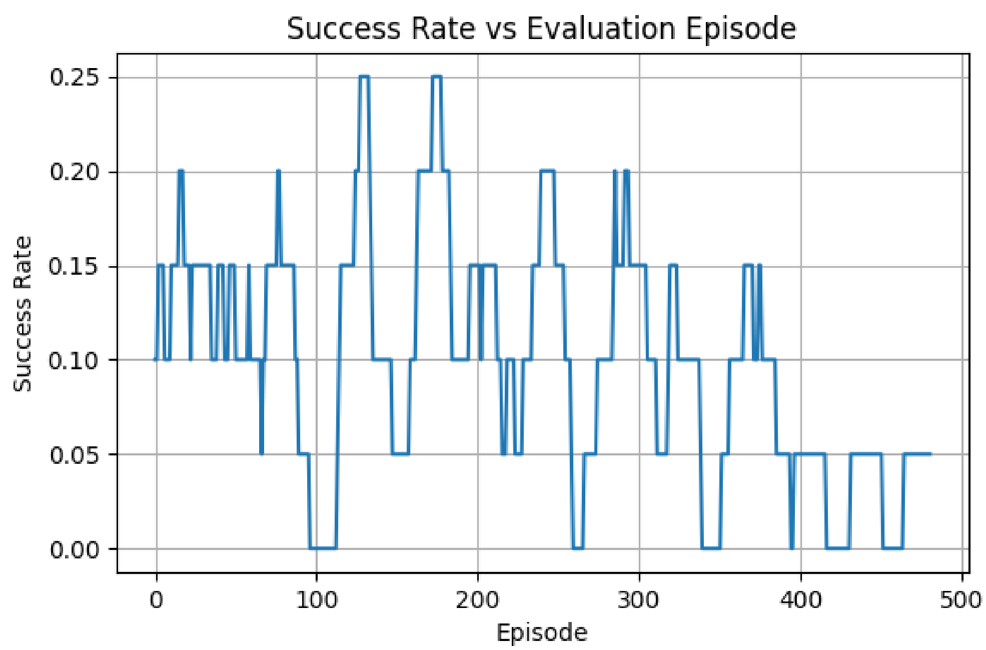

Fig. 4: The evaluation success rate indicates a growing capacity to achieve objectives.

V. Conclusion

This work offers a logical combination of learning-driven decision-making and traditional control theory to provide stable and smooth robotic movements. The proposed DAQ-L paradigm combines diffusion-inspired action priors with value-based RL to allow directed exploration while maintaining trajectory viability. Moreover, the QP-based Control Lyapunov Function (CLF) stabilizer provides formal stability guarantees and actuator limitations with minimal impact on the nominal policy behavior. Experimental examination employing a range of complementary metrics, such as average return, success rate, jerk, and trajectory smoothness, shows uniform and steady learning properties. The results show that DAQ-L can generate smooth and physically realistic control trajectories while maintaining task-related performance under realistic assessment situations. Even though the absolute success rates are still moderate because of task impediments, reward systems, and conservative success criteria, the performance patterns demonstrate ongoing policy improvements and increasing task proficiency. Finally, the approach establishes a basic link between stability-constrained control, reinforcement learning, and generative modeling. The results show that DAQ-L is a good choice for robotic inspection and manipulation tasks that call for smooth, safe, and dependable motion performance. In the face of real-world unpredictability, future research will concentrate on ensuring reliable sim-to-real implementation, increasing sampling efficiency, and transitioning to higher-dimensional robotic systems.

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